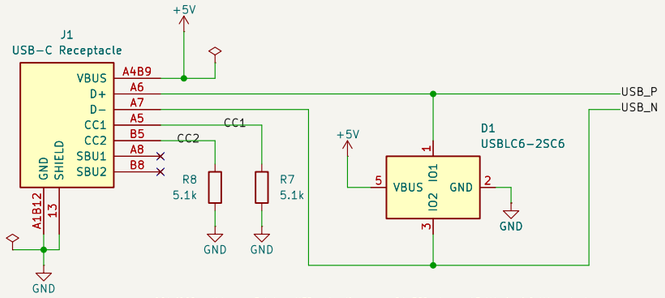
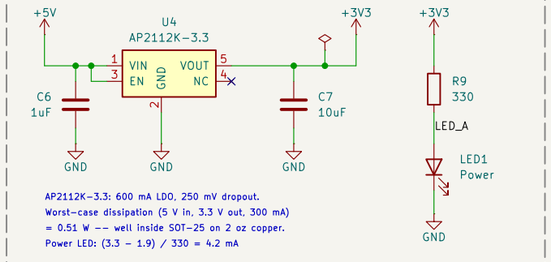
USB-C INPUT & ESD PROTECTION



CC1/CC2 pulldowns: 5.1 k = UFP, advertises a default-current (500 mA) USB 2.0 sink.

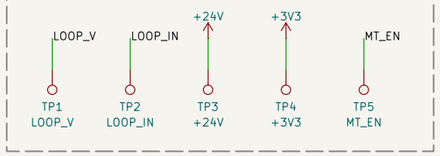
D1 ESD clamp: 15 kV air / 8 kV contact (IEC 61000-4-2). On the board D+ / D- run through its pads: pins 1+6 and 3+4.

3.3 V REGULATOR & POWER LED

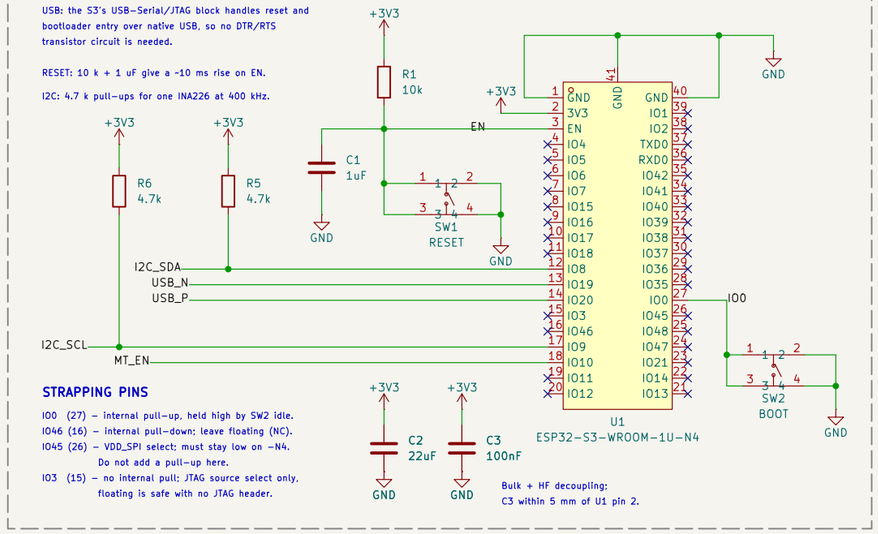


AP2112K-3.3: 600 mA LDO, 250 mV dropout.
Worst-case dissipation (5 V in, 3.3 V out, 300 mA) = 0.51 W -- well inside SOT-25 on 2 oz copper.
Power LED: (3.3 - 1.9) / 330 = 4.2 mA

TEST POINTS



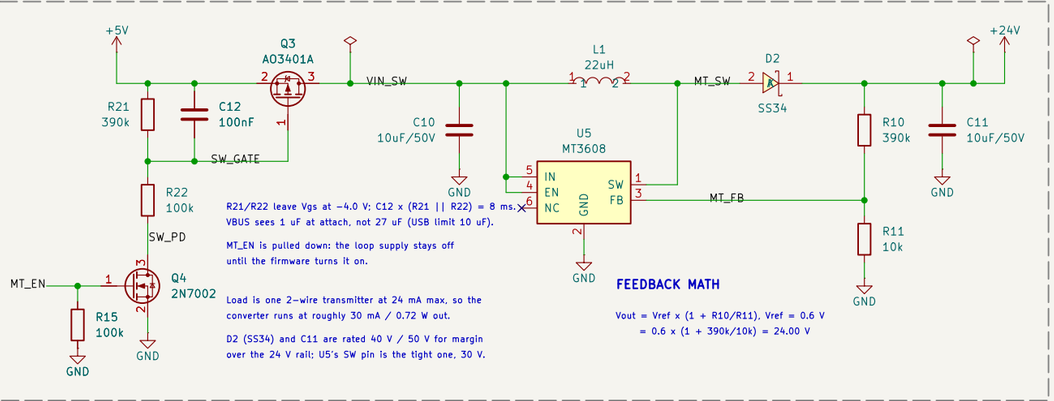
ESP32-S3 MODULE -- reset, boot, I2C pull-ups, decoupling



STRAPPING PINS

IO0 (27) -- internal pull-up, held high by SW2 idle.
IO46 (16) -- internal pull-down; leave floating (NC).
IO45 (26) -- VDD_SPI select; must stay low on -N4. Do not add a pull-up here.
IO3 (15) -- no internal pull; JTAG source select only, floating is safe with no JTAG header.

24 V BOOST CONVERTER -- soft-start load switch, MT3608, feedback



R21/R22 leave Vgs at -4.0 V; C12 x (R21 || R22) = 8 ms. VBUS sees 1 uF at attach, not 27 uF (USB limit 10 uF).

MT_EN is pulled down: the loop supply stays off until the firmware turns it on.

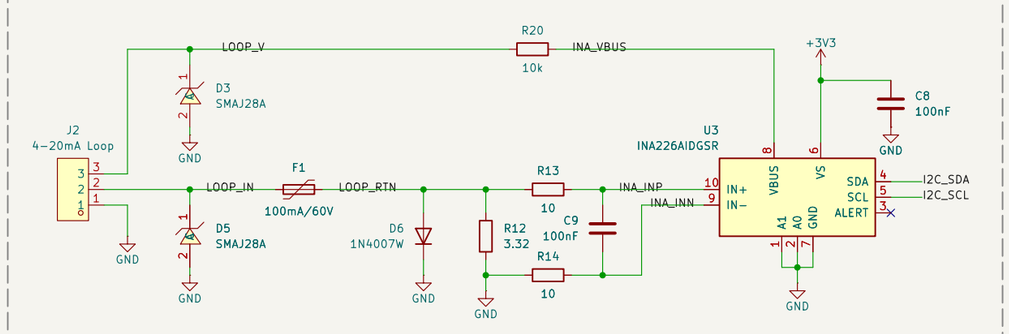
Load is one 2-wire transmitter at 24 mA max, so the converter runs at roughly 30 mA / 0.72 W out.

D2 (SS34) and C11 are rated 40 V / 50 V for margin over the 24 V rail; U5's SW pin is the tight one, 30 V.

FEEDBACK MATH

Vout = Vref x (1 + R10/R11). Vref = 0.6 V
= 0.6 x (1 + 390k/10k) = 24.00 V

LOOP TERMINAL & CURRENT MEASUREMENT -- protection, shunt, filter, INA226



SHUNT SIZING

R12 = 3.32 ohm: the INA226's +/-81.92 mV full scale then reaches 24.67 mA, so a sensor driving 23.5 mA over-range still reads instead of clipping.
24 mA x 3.32 = 79.7 mV (97% of range): LSB 2.5 uV -> 0.75 uA. Dissipation at 24 mA = 1.9 mW (0805).

R13/R14 + C9: 10 ohm / 100 nF differential filter (fc ~ 80 kHz). R14 taps the shunt's own ground pad.

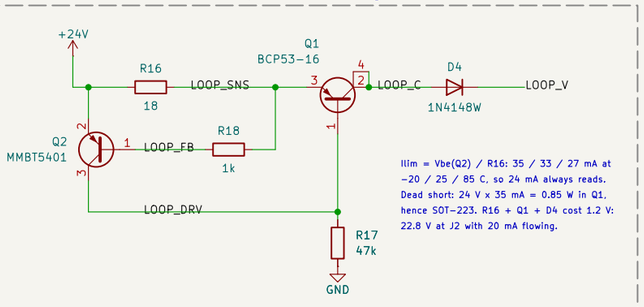
J2: 1 = GND, 2 = mA in, 3 = +24 V out.
2-wire sensor on 3-2; sourcing sensor on 2-1.

PROTECTION

D3 / D5 (28 V) clamp each field wire to ground at 45 V. R20 holds that into U3's 40 V bus pin to 0.5 mA (costs 1.19% of bus-voltage gain).

F1 + D6: with 24 V wired to the mA pin, or in a surge, the shunt sees D6's 1 V, not the fault, and F1 opens it. D6 at 82 mV steals 1 ppm at 25 C, 77 ppm at 85 C.

LOOP CURRENT LIMITER -- 35 mA, reverse blocking



Ilim = Vbe(Q2) / R16: 35 / 33 / 27 mA at -20 / 25 / 85 C, so 24 mA always reads. Dead short: 24 V x 35 mA = 0.85 W in Q1, hence SOT-223. R16 + Q1 + D4 cost 1.2 V: 22.8 V at J2 with 20 mA flowing.

POWER RAILS & SIGNAL LINKS

POWER RAILS (global power ports)

+5V	USB-C VBUS -> LDO, load switch
+3V3	LDO -> ESP32-S3, INA226
+24V	boost -> limiter -> loop terminal
GND	single ground plane

SIGNAL LINKS (net labels)

USB_P / USB_N	USB-C -> ESP32-S3
I2C_SDA / I2C_SCL	ESP32-S3 -> INA226
MT_EN	ESP32-S3 -> load switch
LOOP_V	limiter -> terminal

USB-C powered 4-20 mA loop reader with onboard 24 V loop supply

Sheet: /
File: esp32-4to20ma-board.kicad_sch

Title: ESP32 4-20 mA ESPHome Sensor Board

Size: A3	Date: 2026-11-11	**Rev: H**
KiCad E.D.A. 10.0.6		Id: 1/1